

Multi-Task Graph Neural Networks for Political Account Identification and Misinformation Label Prediction

Political Misinformation Social Network Project

Abstract

This paper investigates multi-task graph neural networks for political communication analysis. A collection of 7,053 tweets from 14 Macedonian political accounts is represented as a graph containing 10,160 nodes and 97,364 edges. A shared graph encoder is trained with two task-specific heads: a 14-class author-identification head and a binary classifier for automatically generated misinformation labels. GCN, GraphSAGE, and GAT are evaluated under a common stratified split. GraphSAGE provides the strongest balance, obtaining an author macro-F1 of 0.989 and a misinformation macro-F1 of 0.708. Single-task controls and training-size sensitivity experiments further show that the joint objective improves misinformation-label prediction for GCN and GraphSAGE, although the effect is architecture-dependent.

Keywords: graph neural networks; social networks; political communication; misinformation; multi-task learning

1. Introduction

Political discussion on social media is shaped by both message content and the network structure through which messages are produced and circulated. Treating tweets as independent observations discards information about authorship, interaction, and shared social context. Graph neural networks (GNNs) provide a framework for combining these sources by propagating feature representations over a graph. This study examines a multi-task formulation of political tweet analysis. The first task is author identification, while the second predicts whether a tweet received a generated misinformation label. The tasks share a graph encoder but use separate output heads. In addition to the main architecture comparison, single-task controls and training-size sensitivity tests are included.

2. Related Work

Graph convolutional networks perform localized neighborhood aggregation for semi-supervised node classification. GraphSAGE learns an aggregation function that can be applied to previously unseen nodes, while graph attention networks use learned attention coefficients to weight neighboring representations. This work applies these architectures to political social-network data while treating misinformation prediction as weak supervision because its labels are generated by a zero-shot language model.

3. Data and Graph Representation

The archive contains 7,053 tweets associated with 14 political accounts. Each tweet is represented using text-derived and metadata features prepared by the preprocessing workflow. Social-network entities and relations extracted from the archive form a graph represented in PyTorch Geometric. The graph contains 10,160 nodes and 97,364 edges. Tweet nodes occur first in the feature matrix and are the only labeled examples; the remaining nodes provide neighborhood context during message passing. Misinformation labels were generated using joeddav/xlm-roberta-large-xnli with candidate labels valid, misleading, and invalid. The class distribution is 5,303 valid, 1,084 invalid, and 666 misleading tweets. These are pseudo-labels rather than expert fact-checks.

4. Methodology

Each model consists of two graph convolution layers with hidden dimension 96. Each layer is followed by layer normalization, ReLU, and dropout. The shared representation feeds a 14-class author head and a binary misinformation head. The objective is the sum of the two cross-entropy losses, with the positive misinformation class weighted by 2.0. GCN, GraphSAGE, and GAT use identical optimization settings and a fixed random seed. Tweet nodes are divided into training, validation, and test partitions using an 80/20 test split followed by an 80/20 training-validation split. Stratification is performed using author labels, and the same partition is reused for all main comparisons. Accuracy and macro-F1 are reported for each task.

5. Main Results

Model	Auth. acc.	Auth. F1	Misinfo acc.	Misinfo F1
GCN	0.962	0.957	0.755	0.689
GraphSAGE	0.991	0.989	0.775	0.708
GAT	0.997	0.997	0.703	0.641

All architectures identify authors more accurately than they predict generated misinformation labels. GAT obtains the highest author score, whereas GraphSAGE obtains the highest misinformation score and the strongest combined performance.

6. Additional Experiments

To separate architectural effects from joint training, each backbone was also trained with only one prediction head. The same graph, split, optimizer, and early-stopping rule were used. The single-task macro-F1 values were: GCN, 0.952 author and 0.683 misinformation; GraphSAGE, 0.998 and 0.693; and GAT, 0.994 and 0.700. Relative to these controls, multi-task training raises misinformation macro-F1 for GCN from 0.683 to 0.689 and GraphSAGE from 0.693 to 0.708. GAT decreases from 0.700 to 0.641 while its author macro-F1 increases from 0.994 to 0.997, indicating that the auxiliary objective can help or compete depending on the encoder.

GraphSAGE training-size sensitivity

Train fraction	Author F1	Misinfo F1
50%	0.990	0.689
75%	0.990	0.712
100%	0.989	0.708

With the validation and test sets fixed, author macro-F1 remains approximately constant while misinformation macro-F1 varies modestly across training fractions. The 100% condition is the main experiment.

7. Discussion

The experiments support multi-task graph learning as a compact way to model political social-network data. The shared representation distinguishes the 14 accounts while supporting a second, weaker classification objective. Architectural choices affect the tasks differently: attention improves author identification in this setting, whereas neighborhood aggregation provides better performance on the misinformation-label task. The generated labels may reflect linguistic or topical patterns rather than verified misinformation, and the random split may place similar content from the same account in both training and test sets.

8. Limitations and Future Work

The principal limitation is the absence of independently verified misinformation annotations. Future work should replace generated

labels with fact-checking labels or annotate a carefully sampled evaluation set. A temporal split would provide a more realistic estimate of generalization to later political events. Further experiments could compare heterogeneous graph formulations, tune task-loss weights, and report calibration and per-class confusion matrices.

9. Conclusion

This paper presented a multi-task GNN framework for political tweet analysis and extended the main comparison with single-task controls and a training-size sensitivity analysis. GraphSAGE achieved the best balance between political-account identification and prediction of generated misinformation labels, while GAT achieved the strongest author-identification results. The findings demonstrate the value of graph representation learning for social-media analysis while emphasizing that weakly supervised misinformation prediction must be interpreted cautiously.

Reproducibility

The preprocessing notebooks, model implementation, dependency list, and Google Colab workflow are available at github.com/VikiShmiki/Political-Misinformation-Social-Networks.

References

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